Johannes Krotz

Curriculum Vitae

✉ *jkrotz@nd.edu*

Applied and computational mathematics with interests in kinetic transport, uncertainty quantification, data assimilation, and scientific machine learning.

Education

2024 **PhD in Mathematics**, *University of Tennessee*, Knoxville, TN, Minor in Computational Sciences.
Dissertation: *Probabilistic and data-driven methods in numerical PDE*

2024 **Master of Science in Statistics**, *University of Tennessee*, Knoxville, TN.

2021 **Master of Science in Mathematics**, *Oregon State University*, Corvallis, OR.

2019 **Master of Science in Physics**, *University of Konstanz*, Konstanz, Germany, With honors.
Thesis: *Computer simulations of gel formation in colloidal systems of sticky rods*

2018 **Bachelor of Science in Mathematics**, *University of Konstanz*, Konstanz, Germany.
Thesis: *Gibt es rotationssymmetrische Fische?* Existence proof for fish-shaped networks evolving under Gaussian curvature flow

2015 **Bachelor of Science in Physics**, *University of Konstanz*, Konstanz, Germany.
Thesis: *Quantum transport in topological insulator-superconductor heterostructures*; conducted at **Reykjavik University, Reykjavik, Iceland**

2012 **Abitur (higher education entrance qualification)**, *Friedrich-Woehler Gymnasium*, Singen, Germany.

Honors, Awards, and Grants

2025 Elected Full Member, Sigma Xi, The Scientific Research Honor Society

2024–2025 Travel support for conference and workshop participation, including IMSI events, ICERM events, and MCM 2025.

2024 Third place, UTK SIAM Research Showcase

2024 Inducted Member, The Honor Society of Phi Kappa Phi

2023 First place, UTK SIAM Research Showcase

2023 Academic Excellence Fellowship, UTK Mathematics

2023 UTK Graduate Student Senate Travel Award

2022 Randall E. Cline Award

2022 National Science Foundation Mathematical Sciences Graduate Internship (NSF MSGI)

2022 Academic Excellence Fellowship, UTK Mathematics

2021 F. M. Dryzer Memorial Graduate Fellowship

2021 National Science Foundation Mathematical Sciences Graduate Internship (NSF MSGI)

2020 National Science Foundation Mathematical Sciences Graduate Internship (NSF MSGI)

2018 Second place, LöPhi Award for excellent teaching, awarded through the physics student union according to student vote

Research and Professional Experience

2024–present **Postdoctoral Researcher, University of Notre Dame, Department of Aerospace and Mechanical Engineering**. Developing Monte Carlo, hybrid deterministic, and scientific machine learning methods for kinetic and radiation transport, with recent work on generative particle transport, neural moment closures, nonlinear transport algorithms, and scalable hybrid methods. Collaborating with the Center for Advancing the Radiation Resilience of Electronics and the Center for Exascale Monte Carlo Neutron Transport.

2023 **Graduate Research Assistant, ORNL-funded project**. Developed dynamic-likelihood filtering methods for stochastic advection-diffusion equations with Juan Restrepo and Jorge Ramirez.

2021–2022 **Research Intern, ORNL – Mathematics in Computation, funded through NSF MSGI**. Developed hybrid algorithms for simulating complex particle systems in 1D, 2D, and 3D with Cory Hauck and Ryan McClarren.

2020 **Research Intern, LANL – Computational Earth Science, funded through NSF MSGI**. Developed fast 2D and 3D Poisson-disk sampling algorithms for high-quality triangulations of discrete fracture networks with Carl Gable, Jeffrey Hyman, and Matthew Sweeney.

2019 **Master's Student, AG Nielaba – Statistical and Computational Physics, University of Konstanz**. Simulated colloidal systems and anisotropic particles, including clustering, topological data analysis, and shear-stress analysis.

2018 **Bachelor's Student, AG Schnuerer – Differential Geometry, University of Konstanz**. Studied existence problems for Gaussian and mean curvature flows under symmetry constraints.

2015 **Exchange Scholar, Reykjavik University; Bachelor's Student, AG Belzig – Quantum Transport**. Conducted theoretical work on topological insulator-superconductor systems with Sigurður Ingi Erlingsson.

Publications

1 *Generative Monte Carlo Sampling for Constant-Cost Particle Transport*, Joseph Farmer, Aidan Murray, **Johannes Krotz**, Ryan G. McClarren, accepted for publication at *PNAS*.

2 *Labeled-particle hybrid Monte Carlo–discrete ordinates transport with quasi-Monte Carlo sampling*, **Johannes Krotz**, Ryan G. McClarren, accepted and to be published in *Nuclear Science and Engineering*.

3 *Statistical Models for Generative Monte Carlo Particle Transport*, Joseph Farmer, **Johannes Krotz**, Ryan G. McClarren, Todd S. Palmer, submitted to International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering 2027.

4 *Reversible Computing for Monte Carlo Particle Transport*, Aubrey M. Denico, **Johannes Krotz**, Ryan G. McClarren, submitted to International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering 2027.

5 *Recovering Independent Monte Carlo Histories in Nonlinear Boltzmann Transport*, **Johannes Krotz**, Ryan G. McClarren, submitted to International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering 2027.

6 *A Nonlinear, 3 Temperature Model for Non-Grey Radiative Transfer*, **Johannes Krotz**, Ryan G. McClarren, under review at *SIAM Journal on Applied Mathematics*.

7 *Energy landscape and dissipation of sliding magnetic rotor arrays*, **Johannes Krotz**, Anton Lüders, under review at *Physical Review E*.

8 *A Hyperbolic Neural Closure for M1 Radiation Transfer*, Bongseok Kim, Jiahao Zhang, **Johannes Krotz**, Dinshaw Balsara, Ryan McClarren, Guang Lin, under review at *Computer Methods in Applied Mechanics and Engineering*.

9 *Scatter-Limited Hybrid Monte Carlo, Deterministic Transport with Quasi–Monte Carlo Sampling*, **Johannes Krotz**, Ryan G. McClarren, in *Proceedings of PHYSOR 2026*.

10 *Curvilinear coordinates and curvature in radiative transport*, **Johannes Krotz**, Ryan G. McClarren, arXiv:2508.20852; submitted Aug. 28, 2025; under review at *Journal of Computational and Theoretical Transport*.

11 *A dynamic likelihood approach to filtering transport processes: advection-diffusion dynamics*, **Johannes Krotz**, Juan M. Restrepo, Jorge Ramirez, *Journal of Computational Physics*, Volume 536 (2025).

12 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, **Johannes Krotz**, Cory D. Hauck, Ryan G. McClarren, *Journal of Computational Physics*, Volume 514 (2024).

13 *Variable resolution Poisson-disk sampling for meshing discrete fracture networks*, **Johannes Krotz**, Matthew R. Sweeney, Carl W. Gable, Jeffrey D. Hyman, Juan M. Restrepo, *Journal of Computational and Applied Mathematics*, Volume 407 (2022).

Presentations

Highlighted Talks

2026 *Labeled-particle hybrid Monte Carlo–discrete ordinates transport with quasi-Monte Carlo sampling*, presented at SIAM Annual Meeting 2026, Cleveland, OH, USA

2026 *Scatter-Limited Hybrid Monte Carlo, Deterministic Transport with Quasi–Monte Carlo Sampling*, presented at PHYSOR 2026, Turin, Italy

2026 *Generative Models for Physical Processes*, presented at the 2026 MICDE Predictive Science Symposium, Ann Arbor, MI

2025 *Normalizing Flow-Based Monte Carlo Transport Simulation*, presented at the International Conference on Monte Carlo Methods and Applications (MCM 2025), Chicago, IL, USA

2025 *Hybrid Monte Carlo Methods for Kinetic Transport*, presented at the International Conference on Monte Carlo Methods and Applications (MCM 2025), Chicago, IL, USA

2025 *Minimizing Uncertainty in Transport Problems through Dynamic Likelihood Filtering*, presented at the Southeastern Atlantic Section of the SIAM Section Meeting 2025, Knoxville, TN

2024 *Hybrid Methods for Kinetic Transport Equations*, presented at the SIAM New York–New Jersey–Pennsylvania Section Conference, Rochester, NY, USA

2023 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, presented at the 6th Annual SIAM Texas–Louisiana Sectional Meeting, Lafayette, LA

Other Talks

2025 *Computational Hybrid Methods for Kinetic Transport*, presented at the COSE Research Horizons Symposium, University of Notre Dame

2025 *MCDG for MCDC – Hybrid Monte Carlo Discontinuous Galerkin Methods for Exascale Transport*, presented at the 2025 PSAAP III Final Review Meeting, Raleigh, NC

2025 *Minimizing Uncertainty in Transport Problems through Dynamic Likelihood Filtering*, presented at the 2025 AMS Spring Central Sectional Meeting, Lawrence, KS

2025 *Introduction to Bayesian Filtering and Applications in Transport Problems*, presented at the Applied Math and Computation Seminar, Oregon State University

2024 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, presented at the Center for Exascale Monte Carlo Neutron Transport

2024 *A Likelihood Approach to Filtering for Advection Diffusion Processes*, presented at the 103rd Annual MAA Southeastern Section Meeting, Knoxville, TN

2024 *Efficient solvers for kinetic transport equations*, presented at the University of Konstanz, Colloquium on Numerical Optimization

2023 *Efficient solvers for the neutron transport equation*, presented at Karlsruher Institut für Technologie (KIT), Oberseminar (Comp. Science and Mathematical Methods)

2023 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, presented at Summer School on Computational & Data Science at Duke University

2020 *Dataset reduction in TDA with an example from soft matter physics*, presented at the Data Science and Image Analysis Conference of the Pacific Northwest, Pullman, WA

Posters

2026 *And Stay out! And Stay Out! Constant Escape-time GMC* , presented at CARRE: 2026 Alliance Strategy Team Annual Peer Review Meeting

2026 *Generative Monte Carlo for Constant-Cost Particle Transport*, presented at Scientific Machine Learning: Theory, Algorithms, Applications, and Agentic AI Workflows (SciML@ND26), University of Notre Dame

2025 *Dynamic Likelihood Filtering – Data Assimilation for Advective Transport*, presented at Data Assimilation and Inverse Problems for Digital Twins (IMSI/Illinois Institute of Technology)

2025 *Conditional Normalizing Flows for Transport Simulation*, presented at Statistical and Computational Challenges in Probabilistic Scientific Machine Learning (SciML), Chicago

2025 *Minimizing Uncertainty in Transport Problems through Dynamic Likelihood Filtering*, presented at Uncertainty Quantification and Machine Learning for Complex Physical Systems (IMSI/Illinois Institute of Technology)

2025 *A Dynamic Likelihood Approach to Filtering*, presented at the 2025 PSAAP III Final Review Meeting, Raleigh, NC

2024 *A Dynamic Likelihood Approach to Filtering for Advection-Diffusion Dynamics*, presented at Robust Optimization and Simulation of Complex Stochastic Systems (ICERM), Providence, RI

2023 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, presented at the University of Chicago AI+Science Summer School, Chicago, IL

2022 *Hybrid methods for the numerical simulation of complex particle systems*, presented at 2022 NSF MSGI Summer Research Symposium

2019 *Gel formation in colloidal systems of sticky rods*, SFB 1214: Anisotropic Particles as Building Blocks, Konstanz, Germany

2018 *Computer simulation of anisotropic colloidal particles*, Symposium on Anisotropic Particles, Konstanz, Germany

Service

Conference and Professional Service

2026 Co-organizer of the COSE-Horizons symposium at the University of Notre Dame

2026 Organizer of a minisymposium on *Stochastic and Data-Driven Computation in Science and Engineering* at the SIAM Annual Meeting 2026

2026 Panelist for the "Session on AI and Machine Learning" at the 2026 MICDE Predictive Science Symposium

2025 Organizer of a minisymposium on *Applications and Algorithms for Differential Equations in Modeling and Engineering* at the Southeastern Atlantic Section of the SIAM Section Meeting 2025

Editorial and Reviewing

2026 Reviewer for *International Journal for Uncertainty Quantification*

2026 Reviewer for several papers in the *Journal of Scientific Computing*

2026 Reviewer for *Nonlinear Processes in Geophysics*

2025 Guest Editor for a special issue of *Journal of Computational and Theoretical Transport*

2025 Reviewer for several papers in *Journal of Computational and Theoretical Transport*

2025 Reviewer for *Journal of Sound and Vibration*

2025 CRC Press (Taylor & Francis Group) — Book proposal reviewer

Academic and Community Service

2026 Panelist, Career Panel, University of Notre Dame SIAM Student Chapter

2025–2026 Judge, Northern Indiana Regional Science and Engineering Fair

2025 Judge, COSE Research Horizons Symposium

2023–2024 Elected member of UTK's Math Graduate Student Council (MGSCI)

2021–2022 Elected member of UTK's Graduate Student Senate

Community Leadership

2010–2020 Elected board member, Musikverein Bohlingen (orchestra)

- **2nd Chairman, 2018–2020**: Coordinated concerts, fundraisers, festivals, and community events, including events with revenues of up to approximately $300,000.
- **Youth Coordinator, 2013–2018**: Coordinated the youth orchestra, recruited instrumental teachers and conductors, organized the integration of two neighboring youth orchestras, and planned concerts and retreats.

Professional Memberships

Society for Industrial and Applied Mathematics (SIAM); American Mathematical Society (AMS); Association for Women in Mathematics (AWM); American Association for the Advancement of Science (AAAS); SACNAS.

Programming & Computational Skills

Scientific computing: Python, NumPy/SciPy, Numba, MATLAB, C/C++, Fortran; development of Monte Carlo and deterministic transport solvers, stochastic filtering algorithms, mesh-generation tools, and scientific machine learning pipelines.

HPC and tooling	High-performance computing workflows, Git, Bash, OpenMP, LaTeX, Beamer, TikZ, Matplotlib, gnuplot.
Code contributions	Contributed to `MCDC`, a CUDA-compatible exascale neutron transport code using `Numba`; developed a mesh-generation module used in `DFNWorks`.

Languages

German	**Native**	
English	**Fluent**	*7+ years of academic work in the US; 115/120 on TOEFL in 2019*
Italian	**Conversational**	*5 years of instruction; functional with immersion*
Latin	**Reading knowledge**	*Grosses Latinum; not used professionally*

Teaching

Instructor of record where indicated; otherwise teaching assistant.

University of Notre Dame

2024–present	Co-advisor in weekly meetings, guiding the research of Ph.D. students and undergraduate students.

University of Tennessee, Knoxville

Spring 2023	Instructor of Record for Finite Math (MTH 123)
Fall 2022	Instructor of Record for Finite Math (MTH 123)
Spring 2022	Instructor of Record for Basic Calculus (MTH 125)
Fall 2021	Basic Calculus (MTH 125)

Oregon State University

Spring 2021	Advanced Calculus II (MTH 312)
Winter 2021	Advanced Calculus I (MTH 311)
Fall 2020	Linear Algebra II (MTH 342)
Spring 2020	Linear Algebra I (MTH 341)
Winter 2020	College Algebra (MTH 111)
Fall 2019	Elementary Functions (MTH 112)

University of Konstanz

Summer 2019	Real Analysis II (Department of Mathematics)
Summer 2019	Linear Algebra and Analysis for Computer Scientists (Department of Computer and Information Science)
Winter 2018/19	Theoretical Physics III (Department of Physics)
Winter 2018/19	Mathematics for Physicists III (Department of Mathematics)
Summer 2018	Discrete Mathematics (Department of Computer and Information Science)
Summer 2018	Mathematics for Physicists II (Department of Mathematics)
Winter 2017/18	Mathematics for Physicists III (Department of Mathematics)
Summer 2017	Mathematics for Physicists II (Department of Mathematics)
Winter 2016/17	Mathematics for Physicists I (Department of Mathematics)